Q If in an AP, $a_m = \frac{1}{n}$ and $a_n = \frac{1}{m}$, then find $a_{m+n} = ?$

Q Find sum of first 24 terms of an AP, if it is known that

$$a_1 + a_5 + a_{10} + a_{15} + a_{20} + a_{24} = 225.$$

Sol^n $a_m = \frac{1}{n}$ and $a_n = \frac{1}{m}$

$a + (m-1)d = \frac{1}{n}$ — (1) | $a + (n-1)d = \frac{1}{m}$ — (2)

(1) − (2)

$\{a + md - d\} - \{a + nd - d\} = \frac{1}{n} - \frac{1}{m}$

$\cancel{a} + md - \cancel{d} - \cancel{a} - nd + \cancel{d} = \frac{m-n}{mn}$

$(m-n)d = \frac{m-n}{mn}$

$d = \frac{1}{mn}$

$d = \frac{1}{mn}$ in eq (1)

$a + \frac{(m-1)}{mn} = \frac{1}{n}$

$a = \frac{1}{n} - \frac{m-1}{mn}$

$= \frac{m-(m-1)}{mn}$

$= \frac{1}{mn}$

$a_{m+n} = a + (m+n-1)d$

$= \frac{1}{mn} + \frac{m+n-1}{mn}$

$= \frac{1+m+n-1}{mn}$

$= \frac{m+n}{mn}$

Q $a_1, a_2, a_3 \, - - \, \_\_\_\_ , a_{2n}$ are in AP, then find value of

$$a_1^2 - a_2^2 + a_3^2 - a_4^2 + \cdots\cdots + a_{2n-1}^2 - a_{2n}^2$$ in terms of a_1, a_{2n} and n.

Q If $a_1, a_2, a_3 \, \cdots\cdots , a_n$ are in AP, then find

$$\frac{1}{\sqrt{a_1}+\sqrt{a_2}} + \frac{1}{\sqrt{a_2}+\sqrt{a_3}} + \frac{1}{\sqrt{a_3}+\sqrt{a_4}} + \cdots\cdots + \frac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}} = ?$$

(in terms of a_1, a_n and n)

$a_1 + a_5 + a_{10} + a_{15} + a_{20} + a_{24} = 225$

$a + a + 4d + a + 9d + a + 14d + a + 19d + a + 23d = 225$

$6a + 69d = 225$

$2a + 23d = 75$

$S_{24} = \frac{24}{2}\left[2a + (24-1)d\right]$

$= 12\left[2a + 23d\right]$

$= 12 \times 75$

$= 900$

$a_1, a_2, a_3 - - - - - - - - a_{2n}$ are in AP

$a_2 - a_1 = a_4 - a_3 = a_6 - a_5 = - - \text{---} = d$

$$(a_1^2 - a_2^2) + (a_3^2 - a_4^2) + (a_5^2 - a_6^2) + - - - - - - + (a_{2n-1}^2 - a_n^2)$$

$$= (a_1 + a_2)(a_1 - a_2) + (a_3 + a_4)(a_3 - a_4) + (a_5 + a_6)(a_5 - a_6) + - - - - - - + (a_{2n-1} + a_{2n})(a_{2n-1} - a_{2n})$$

$$= (a_1 + a_2)(-d) + (a_3 + a_4)(-d) + (a_5 + a_6)(-d) + - - - - - - + (a_{2n-1} + a_{2n})(-d)$$

$$= (-d)\left[a_1 + a_2 + a_3 + a_4 + a_5 + a_6 + - - - - - - + a_{2n-1} + a_{2n}\right]$$

$$= (-d)\,\frac{\cancel{2}n}{\cancel{2}}\left[a_1 + a_{2n}\right]$$

$$= -\left(\frac{a_{2n} - a_1}{2n-1}\right) n \left[a_{2n} + a_1\right]$$

$$= \frac{n}{1-2n}\left(a_{2n}^2 - a_1^2\right)$$

$$a_{2n} = a_1 + (2n-1)d$$

$$a_{2n} - a_1 = (2n-1)d$$

$$\therefore d = \frac{a_{2n} - a_1}{2n-1}$$

$$\frac{1}{\sqrt{a_1}+\sqrt{a_2}}+\frac{1}{\sqrt{a_2}+\sqrt{a_3}}+\frac{1}{\sqrt{a_3}+\sqrt{a_4}}+\cdots+\frac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}}$$

$$=\frac{\sqrt{a_1}-\sqrt{a_2}}{a_1-a_2}+\frac{\sqrt{a_2}-\sqrt{a_3}}{a_2-a_3}+\frac{\sqrt{a_3}-\sqrt{a_4}}{a_3-a_4}+\cdots+\frac{\sqrt{a_{n-1}}-\sqrt{a_n}}{a_{n-1}-a_n}$$

$$=\frac{\sqrt{a_1}-\sqrt{a_2}}{-d}+\frac{\sqrt{a_2}-\sqrt{a_3}}{-d}+\frac{\sqrt{a_3}-\sqrt{a_4}}{-d}+\cdots+\frac{\sqrt{a_{n-1}}-\sqrt{a_n}}{-d}$$

$$=\frac{(\sqrt{a_1}-\cancel{\sqrt{a_2}})+(\cancel{\sqrt{a_2}}-\cancel{\sqrt{a_3}})+(\cancel{\sqrt{a_3}}-\cancel{\sqrt{a_4}})+\cancel{\cdots}+(\cancel{\sqrt{a_{n-1}}}-\sqrt{a_n})}{-d}$$

$$=\frac{\sqrt{a_1}-\sqrt{a_n}}{-d}$$

$$=\frac{\sqrt{a_1}-\sqrt{a_n}}{-\left(\frac{a_n-a_1}{n-1}\right)}=$$

$$=-(n-1)\frac{\sqrt{a_1}-\sqrt{a_n}}{a_n-a_1}$$

$$=(n-1)\frac{\cancel{(\sqrt{a_n}-\sqrt{a_1})}}{\cancel{(\sqrt{a_n}-\sqrt{a_1})}(\sqrt{a_n}+\sqrt{a_1})}=\frac{n-1}{\sqrt{a_n}+\sqrt{a_1}}$$

$$a_n-a_1$$

$$(\sqrt{a_n}-\sqrt{a_1})(\sqrt{a_n}+\sqrt{a_1})$$

$$a_n=a_1+(n-1)d$$

$$d=\frac{a_n-a_1}{n-1}$$

Q Find relation between a and d, if $\frac{S_{2n}}{S_n}$ doesnot depend on values of n,

where (S_n is the sum of the 1st n terms of an AP)

Sol^n

$$\frac{S_{2n}}{S_n} = \frac{\frac{\cancel{2n}}{\cancel{2}}[2a+(2n-1)d]}{\frac{\cancel{n}}{2}[2a+(n-1)d]}$$

$$= \frac{2[2a+2nd-d]}{[2a+nd-d]}$$

$$= 2\,\frac{\boxed{(2a-d)}+2nd}{\boxed{(2a-d)}+nd}$$

$2a-d=0$

$$\boxed{2a=d}$$

Q If sum of n terms of 2 AP's are in the ratio $\frac{5n+4}{6n+5}$, find ratio of their 10th term.

Soln

a_{10}, n, a, d, S_n | n, a', d', S_n', a_{10}'

$$\frac{a_{10}}{a_{10}'} = \boxed{\frac{a+9d}{a'+9d'}} = ?$$

$$\frac{S_n}{S_n'} = \frac{5n+4}{6n+5}$$

$$\Rightarrow \frac{\cancel{\frac{n}{2}}\{2a+(n-1)d\}}{\cancel{\frac{n}{2}}\{2a'+(n-1)d'\}} = \frac{5n+4}{6n+5}$$

$$\Rightarrow \frac{2a+(n-1)d}{2a'+(n-1)d'} = \frac{5n+4}{6n+5}$$

$$\Rightarrow \frac{\frac{2a}{2}+\frac{(n-1)d}{2}}{\frac{2a'}{2}+\frac{(n-1)d'}{2}} = \frac{5n+4}{6n+5}$$

$$\frac{a+\left(\frac{n-1}{2}\right)d}{a'+\left(\frac{n-1}{2}\right)d'} = \frac{5n+4}{6n+5}$$

$\frac{n-1}{2} \to 9$

$n = 19$

$$\frac{a+9d}{a'+9d'} = \frac{5\times19+4}{6\times19+5}$$

$$= \frac{99}{119}$$

Q given 2 AP's → 2, 5, 8, 11, ... (60 terms)

3, 5, 7, 9, ... (50 terms), then find

no of common terms of those two AP.

Sol'n

2, 5, 8, 11, ..., 179

3, 5, 7, 9, ..., 101

$t_{60} = 2 + (60-1)3 = 179$

$b_{50} = 3 + (50-1)2 = 101$

common terms. 5, 11, 17, ..., a_n

$a_n \leq 101$

$5 + (n-1)6 \leq 101$

$(n-1)6 \leq 96$

$n-1 \leq 16$

$n \leq 17$

$n = 17$

if $n \leq 16.4$

then $n = 16$

→ 'cd' of common terms is LCM of given differences

Q A software company sets up 'm' computer systems to finish an assignment in 17 days. If 4 computer systems crashed on start of the 2nd day, 4 more on the start of 3rd day and so on, then it took 8 more days to finish the assignment. Then value of m is equal to

Soln:

'm' computer completes $\frac{1}{17}$th work in a day

1 " " $\frac{1}{17m}$th " " " "

A/q

$$\underbrace{\left[m \times \frac{1}{17m}\right] + \left[(m-4)\frac{1}{17m}\right] + \left[(m-8)\frac{1}{17m}\right] + \cdots}_{25 \text{ days}} = 1$$

$$\frac{1}{17m}\left[m + (m-4) + (m-8) + \cdots\right] = 1$$

$$\frac{1}{17m} \times \frac{25}{2}\left[2m + (25-1)(-4)\right] = 1$$

$$25\left[2m - 96\right] = 34m$$

$$\Rightarrow 50m - 2400 = 34m$$

$$\Rightarrow 16m = 2400$$

$$\Rightarrow m = \frac{2400}{16} = 150$$

6:00pm

Properties of AP

a) If a fixed number is added or subtracted from terms of AP, then it will still be an AP → cd doesnot change

→ If all the terms are multiplied or divided by a fixed number then it will still be an AP and cd will change accordingly

b) $a_1, a_2, a_3 \text{ --- } , a_n$ be an AP , $cd = d_1$

$b_1, b_2, b_3 \text{ --- } , b_n$ be an AP , $cd = d_2$

$a_1+b_1, a_2+b_2, a_3+b_3, \text{ --- }$ will be an AP , $cd = d_1 + d_2$

c) if $a_1, a_2, a_3 \text{ --- } , a_n$ be an AP

then $a_1 + a_n = a_2 + a_{n-1} = a_3 + a_{n-2} = \text{ --- }$


Sticker
2D

d) if 3 numbers are in AP, $a-d, a, a+d \rightarrow cd = d$

if 4 numbers are in AP, $a-3d, a-d, a+d, a+3d \rightarrow cd = 2d$

if 5 " " " ", $a-2d, a-d, a, a+d, a+2d \rightarrow cd = d$

Q Divide 32 into 4 parts which are in AP, such that the ratio of the product of extremes to the product of means is $\frac{7}{15}$. Find those parts.

Soln parts are $\rightarrow a-3d, a-d, a+d, a+3d$

A/Q $\frac{(a-3d)(a+3d)}{(a-d)(a+d)} = \frac{7}{15}$

$\Rightarrow \frac{a^2-9d^2}{a^2-d^2} = \frac{7}{15}$

$\Rightarrow 15a^2 - 135d^2 = 7a^2 - 7d^2$

$\Rightarrow 8a^2 = 128d^2$

$\Rightarrow a^2 = 16d^2$

$\Rightarrow 64 = 16d^2$

$\Rightarrow 4 = d^2$

$d = \pm 2$

A/Q

$(a-\cancel{3d}) + (a-\cancel{d}) + (a+\cancel{d}) + (a+\cancel{3d}) = 32$

$4a = 32$

$a = 8$

$a = 8, d = 2 \Rightarrow 2, 6, 10, 14$

$a = 8, d = -2 \Rightarrow 14, 10, 6, 2$

Geometric Progression

eg 3, 6, 12, 24, 48, 96, ----

32, −16, 8, −4, 2, −1, $\frac{1}{2}$, $\frac{-1}{4}$, ----

$a, ar, ar^2, ar^3, ar^4, \ldots, ar^{n-1}$, $t_n = ar^{n-1}$

$$\frac{t_2}{t_1} = \frac{t_3}{t_2} = \frac{t_4}{t_3} = \ldots = r$$

Q If 4th term and 7th term of a GP be 10 and 80 respectively, then find

a) common ratio

b) first term

c) 13th term

Soln

$t_4 = 10$ | $t_7 = 80$

$ar^3 = 10$ —(1) | $ar^6 = 80$ —(2)

(2) ÷ (1) $\frac{\cancel{a}r^6}{\cancel{a}r^3} = \frac{80}{10}$

$r^3 = 8$ $\boxed{r = 2}$

$r = 2$ in (1)

$a \cdot 2^3 = 10$

$a = \frac{10}{8}$

$\boxed{a = \frac{5}{4}}$

$t_{13} = ar^{12}$

$= \frac{5}{4}\, 2^{12}$

$\boxed{t_{13} = 5 \cdot 2^{10}}$

Q If $t_1, t_2, t_3, \ldots$ form an increasing GP and $\frac{t_1+t_3+t_5}{t_1+t_2+t_3} = 3$, find common ratio (given $t_1 > 0$)

Soln

$$\frac{t_1+t_3+t_5}{t_1+t_2+t_3} = 3$$

$$\frac{a+ar^2+ar^4}{a+ar+ar^2} = 3$$

$$\frac{\cancel{a}(1+r^2+r^4)}{\cancel{a}(1+r+r^2)} = 3$$

$$\Rightarrow \frac{\cancel{(r^2+r+1)}(r^2-r+1)}{\cancel{(1+r+r^2)}} = 3$$

$$\Rightarrow r^2 - r - 2 = 0$$

$$r^2 - 2r + r - 2 = 0$$

$$r(r-2) + 1(r-2) = 0$$

$$(r-2)(r+1) = 0$$

$r = 2$ ✓ OR $r = -1$ ✗

$$x^4 + x^2 + 1$$

$$\underbrace{x^4 + 2x^2 + 1} - x^2$$

$$(x^2+1)^2 - x^2$$

$$(x^2+1+x)(x^2+1-x)$$

$$(x^2+x+1)(x^2-x+1)$$

sum of GP

$$S = a + ar + ar^2 + \cdots + ar^{n-1}$$
$$Sr = ar + ar^2 + ar^3 + \cdots + ar^n$$

$$S - Sr = a + \cdots - ar^n$$

$$S(1-r) = a(1-r^n)$$

$$\boxed{S_n = a\left(\frac{1-r^n}{1-r}\right)}$$

$$S = a\left(\frac{r^n - 1}{r-1}\right)$$

($r > 1$, we this)

Q find S

a) $S = 2 + 6 + 18 + \cdots$ (10 terms)

b) $S = 1 + 2 + 2^2 + \cdots + 2^{n-1}$

$s = 2 + 6 + 18 + \cdots$ (10 terms)

$a = 2, r = 3, n = 10$

$$s = 2 \frac{1-3^{10}}{1-3}$$

$$= \cancel{2} \times \frac{1-3^{10}}{(-\cancel{2})}$$

$$= \frac{3^{10}-1}{1}$$

$$= 3^{10}-1$$

$s = 2^0 + 2^1 + 2^2 + \cdots + 2^{n-1}$

$a = 1, r = 2,$ no of terms $= n$

$$s = 1 \cdot \frac{1-2^n}{1-2}$$

$$= 1 \cdot \frac{1-2^n}{(-1)}$$

$$= 2^n - 1$$

Infinite sum

$$1+2+3+4+ \cdots = \infty$$

$$8+6+4+2+ \cdots = -\infty$$

$$1+2+4+8+16+ \cdots = \infty$$

$$8+4+2+1+ \cdots = 16$$

$17 — 18$ $\left| \dfrac{1}{\infty} = 0 \right|$ 16 1+2

$$8+4+2+1+\frac{1}{2}+\frac{1}{3}+ \cdots = 16$$

infinite GP

$$S_n = a\left(\frac{1-r^n}{1-r}\right)$$

$$S_\infty = a\left(\frac{1-r^\infty}{1-r}\right)$$

$$= a\left(\frac{1-0}{1-r}\right)$$

$$\boxed{S_\infty = \frac{a}{1-r}} \quad -1 < r < 1$$

$r = \frac{1}{2}$

$r = \frac{-1}{2}$

$$\left(\frac{1}{2}\right)^\infty = \frac{1^\infty}{2^\infty} = \frac{1}{2^\infty} = \frac{1}{\infty} = 0$$

$$\left(\frac{-1}{2}\right)^\infty = \frac{(-1)^\infty}{2^\infty} = \frac{\pm 1}{\infty} = 0$$

0.5

0.25

0.125

0.0625

0.00 - - - - -

Q finds

a) $s = 1 + \sin^2\theta + \sin^4\theta + \sin^6\theta + \cdots$

b) $s = \frac{1}{2} + \frac{1}{3} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{2^3} + \frac{1}{3^3} + \cdots$

a) $s = 1 + \sin^2\theta + \sin^4\theta + \cdots$

$a = 1, \quad r = \sin^2\theta$

$$S_\infty = \frac{a}{1-r}$$

$$= \frac{1}{1-\sin^2\theta}$$

$$= \frac{1}{\cos^2\theta}$$

$$= \sec^2\theta$$

b) $s = \frac{1}{2} + \frac{1}{3} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{2^3} + \frac{1}{3^3} + \cdots$

$$= \left(\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots\right) + \left(\frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \cdots\right)$$

$$= \frac{1/2}{1-\frac{1}{2}} + \frac{1/3}{1-\frac{1}{3}}$$

$$= \frac{(1/2)}{(1/2)} + \frac{(1/3)}{(2/3)}$$

$$= 1 + \frac{1}{2} = \frac{3}{2}$$